ECE343 Spring 2025

Objective 5.2:
Gauss' Law



UNITED STATES
AIR FORCE
ACADEMY

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 5 Learning Objectives

5.1 I can apply Coloumb's Law to use direct integration to calculate the electric field due to charge density in varoius geometries, including rectangular, cylindrical, and spherical coordinate systems.

5.2 I can apply the integral and differential forms of Gauss' Law to calculate the electric field due to charge density in various geometries, including rectangular, cylindrical, and spherical coordinate systems.

5.3 I can calculate the electric field from the scalar electric potential in a variety of geometries.

5.4 I can apply the electric field boundary conditions to calculate electric fields on both sides of a material boundary.

Gauss' Law

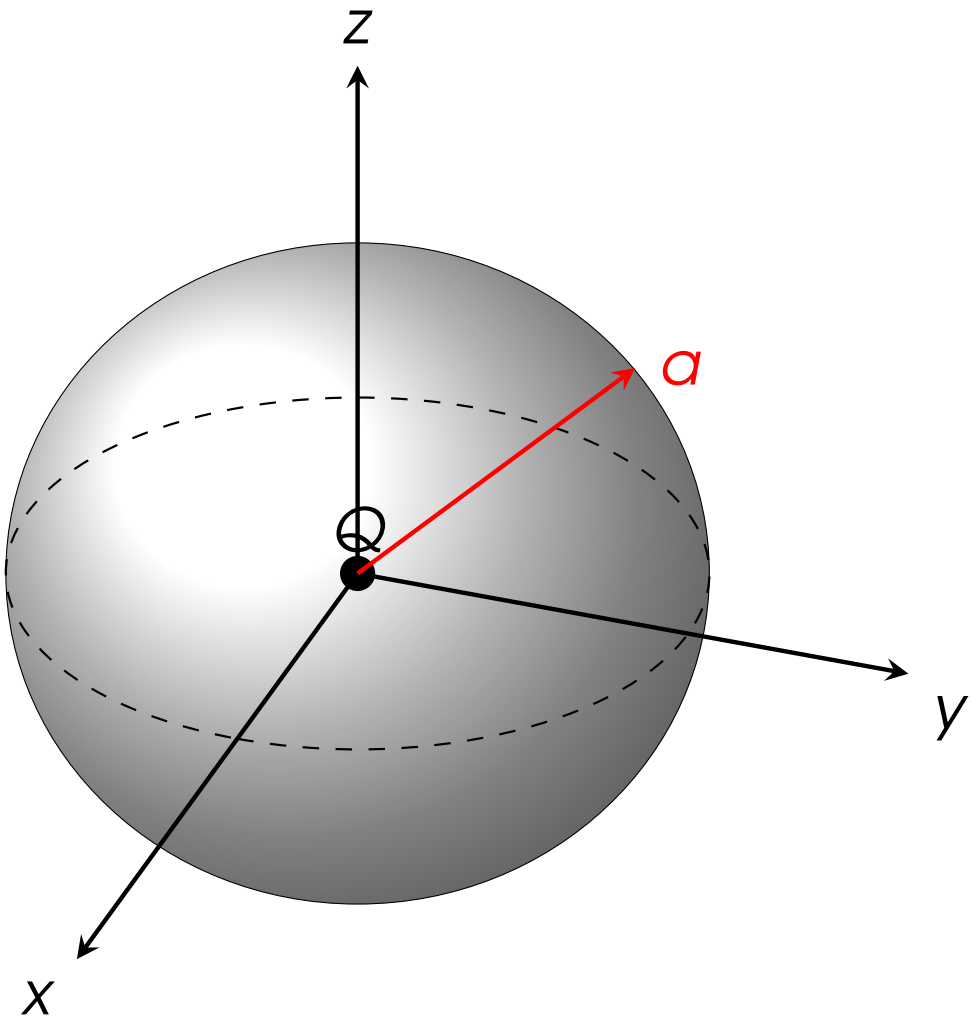
Definition

The flux of the electric charge through a **closed** surface is equal to the enclosed charge.

Integral & Differential (Point) Forms

$$\oiint_S \vec{D} \cdot d\vec{S} = Q_{\text{enclosed}} \qquad \nabla \cdot \vec{D} = \rho_V$$

Gauss' Law: General Method



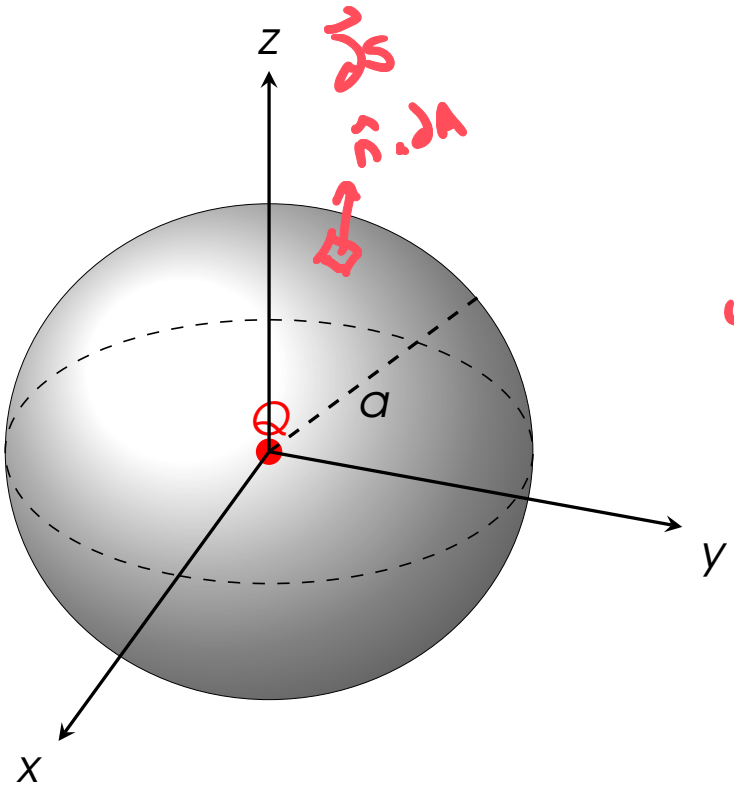
Method

1. Draw Gaussian Surface ($\vec{dS}$)
2. Determine symmetry
3. Solve Left Integral:

$$\oiint_S \varepsilon \vec{E} \cdot \vec{dS}$$

4. Determine charge enclosed, Q_{encl}

GAUSS' LAW: POINT CHARGE



$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q_{encl.}}{\epsilon}$$

$$\int_0^{2\pi}\int_0^{\pi} [\hat{r}E_r + \hat{\theta}E_\theta + \hat{\phi}E_\phi] \cdot \hat{r} r^2 \sin\theta \, d\theta \, d\phi = \frac{Q_{encl.}}{\epsilon}$$

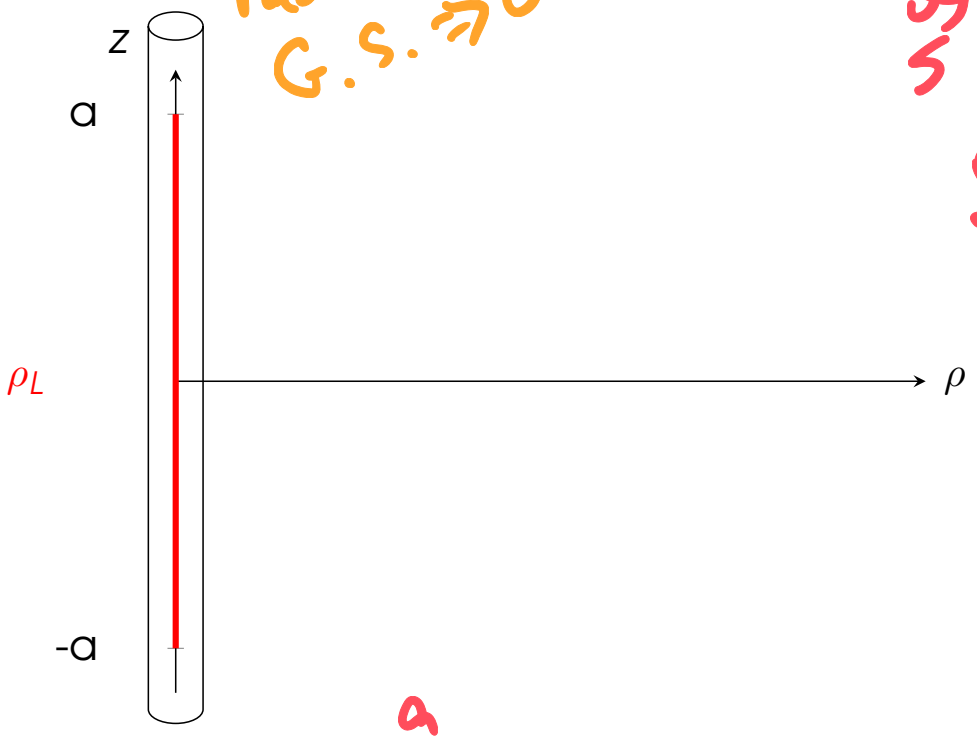
$$\int_0^{2\pi}\int_0^{\pi} E_r r^2 \sin\theta \, d\theta \, d\phi = \frac{Q_{encl.}}{\epsilon}$$

Assume E_r does not depend on θ, ϕ ...

$$E_r r^2 (2\pi)(2) = \frac{Q_{encl.}}{\epsilon}$$

$$\vec{E} = \hat{r}E_r = \hat{r}\frac{Q_{encl.}}{4\pi\epsilon r^2}$$

Gauss' Law: Line Charge



radius of G.S. $\to 0$

$$\oint_S \vec{E}\cdot d\vec{s} = \frac{Q_{encl.}}{\epsilon}$$

$$\int_{\text{Top}} \vec{E}\cdot d\vec{s}_1 + \int_{\text{Bottom}} \vec{E}\cdot d\vec{s}_2 + \int_{\text{Sides}} \vec{E}\cdot d\vec{s}_3 = \frac{Q_{encl.}}{\epsilon}$$

$$\int (\hat{\rho}E_\rho + \hat{\phi}E_\phi + \hat{z}E_z)\cdot\hat{\rho}\rho\,d\phi\,dz$$

$$E_\rho \rho \int_{-a}^{a}\int_{0}^{2\pi} d\phi\,dz = \frac{Q_{encl.}}{\epsilon}$$

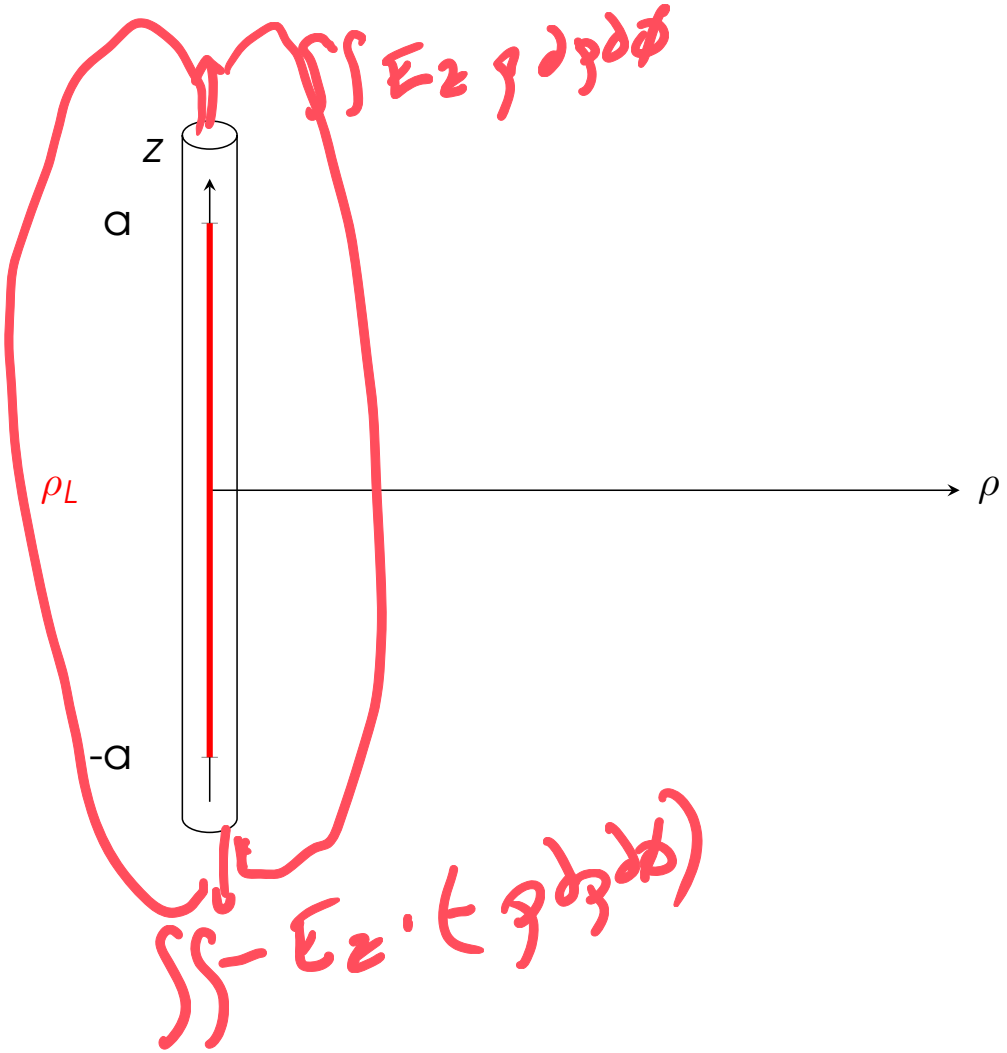
$$E_\rho \rho (2a)(2\pi) = \frac{Q_{encl.}}{\epsilon}$$

$$Q_{encl.} = \int_{-a}^{a} \rho_L\,dz = 2a\rho_L$$

$$\vec{E} = \hat{\rho}E_\rho = \hat{\rho}\frac{Q_{encl.}}{4\pi a\rho\epsilon}$$

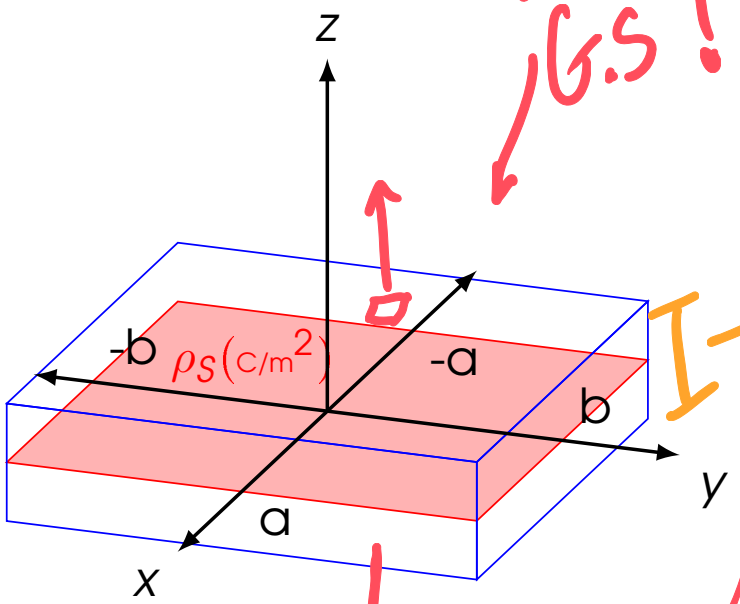
$$\vec{E} = \hat{\rho}\frac{2a\rho_L}{4\pi a\rho\epsilon} = \hat{\rho}\frac{\rho_L}{2\pi\rho\epsilon}$$

Gauss' Law: Line Charge (cont'd)



Gauss' Law: Sheet Charge

Find the Electric field everywhere from a sheet carrying a charge distribution of ρ_S.



another G.S!

$$\iint_{Top} \vec{E}\cdot\vec{ds}_1 + \iint_{Bottom} \vec{E}\cdot ds_2 + \iint_{Sides} \vec{E}\cdot ds_3 = \frac{Q_{encl}}{\epsilon}$$

$I \to 0$

$$Top \to \int_{-b}^{b}\int_{-a}^{a} +\hat{z}E_z\cdot\hat{z}\,dx\,dy = E_z 4ab$$

$$Bottom \to \int_{-b}^{b}\int_{-a}^{a} -\hat{z}E_z\cdot(-\hat{z}\,dx\,dy) = E_z 4ab$$

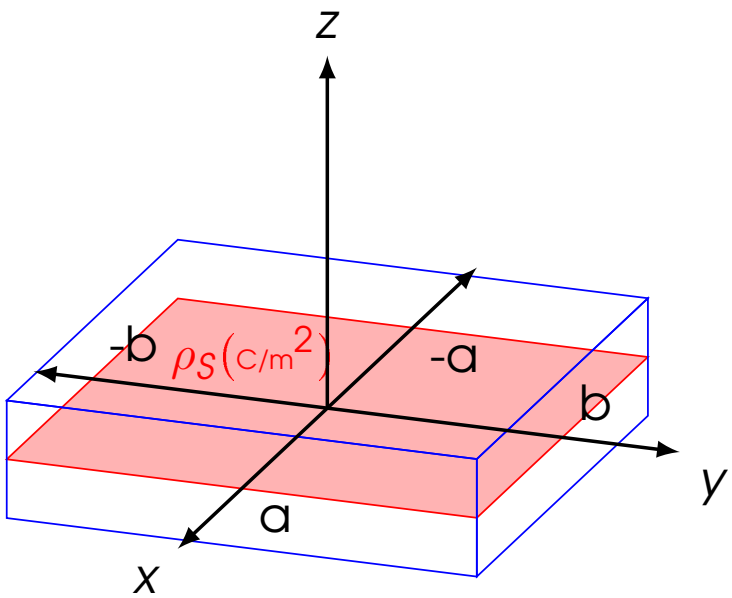
$$Q_{encl} = \int_{-b}^{b}\int_{-a}^{a} \rho_S\,dx\,dy = \rho_S 4ab$$

$$8ab\,E_z = \frac{Q_{encl.}}{\epsilon}$$

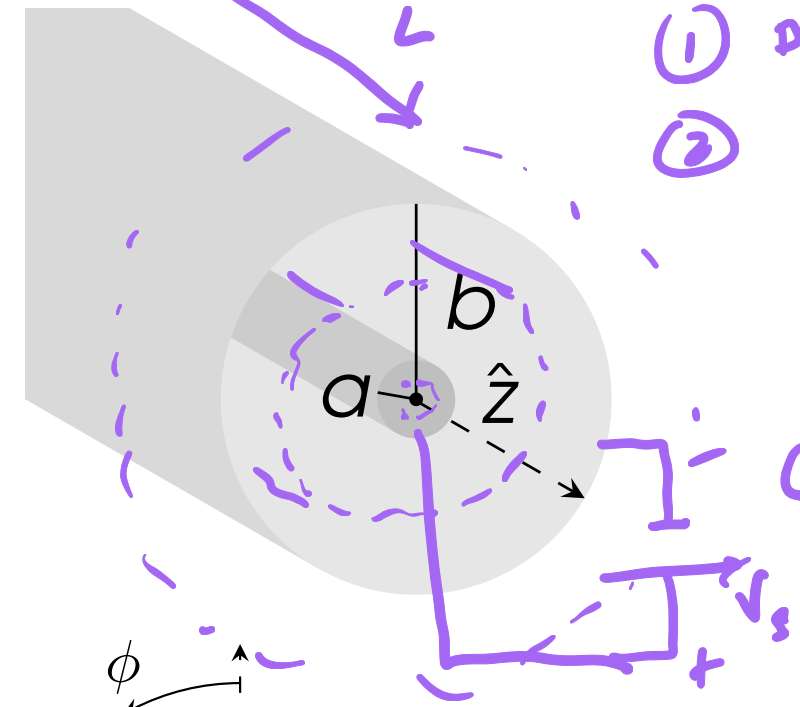
$$\vec{E} = \hat{z}E_z = \hat{z}\frac{\rho_S}{2\epsilon}$$

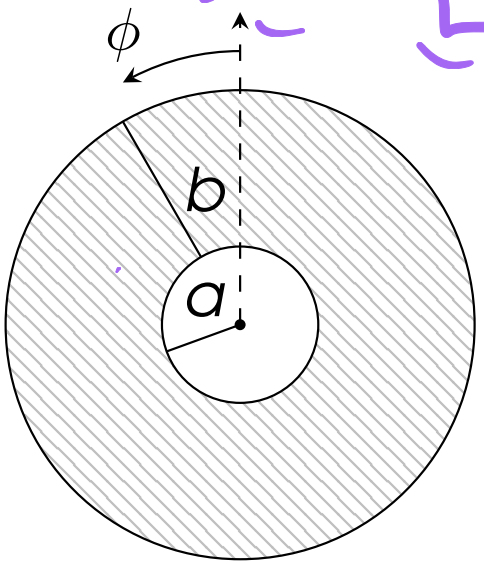
Gauss' Law: Sheet Charge (cont'd)

Find the Electric field everywhere from a sheet carrying a charge distribution of ρ_S.



Gauss' Law: Coaxial Cable — Find $\vec{E}$ everywhere





(1) Draw Picture ✓

(2) Write Gauss' Law

$$\oint_S \vec{E}\cdot d\vec{s} = \frac{Q_{enc.}}{\epsilon}$$

(a) (b)

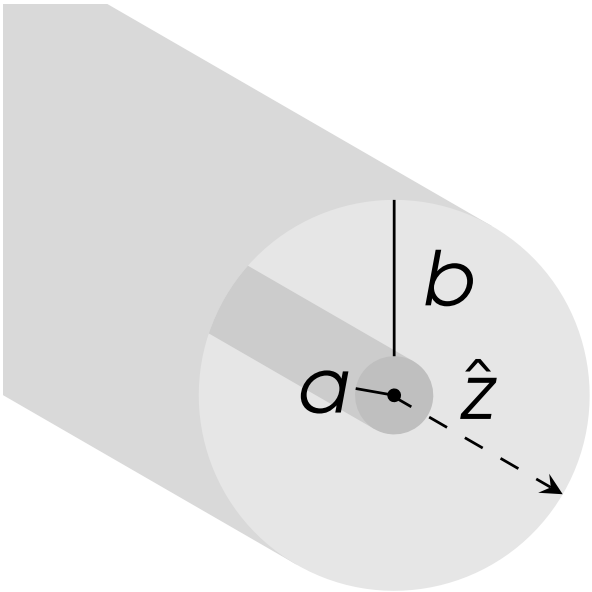
(a) $\vec{E} = \hat{\rho}E_\rho + \hat{\phi}E_\phi + \hat{z}E_z$

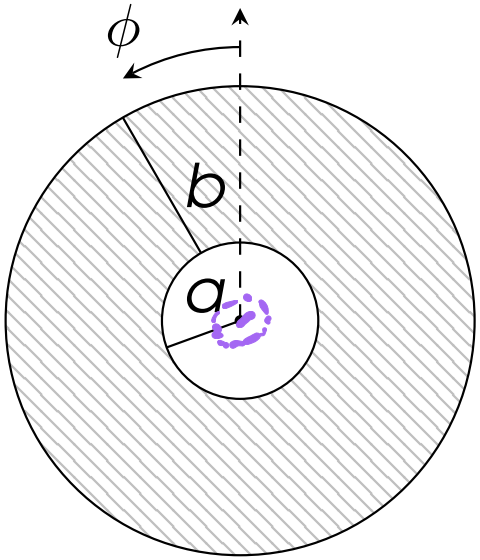
$d\vec{s} = \hat{\rho}\rho\, d\phi\, dz$

$$\int_0^L \int_0^{2\pi} \rho E_\rho \, d\phi \, dz$$

$2\pi\rho L E_\rho$

Gauss' Law: Coaxial Cable (cont'd)





$\rho \rightarrow$ obs. radius

$\rho_0 \rightarrow$ radius associated w/ source

(b) $Q_{encl.} = \iiint \rho_v \, dV$

$\rho_v = \frac{Q_{TOT}}{\pi a^2 L}$

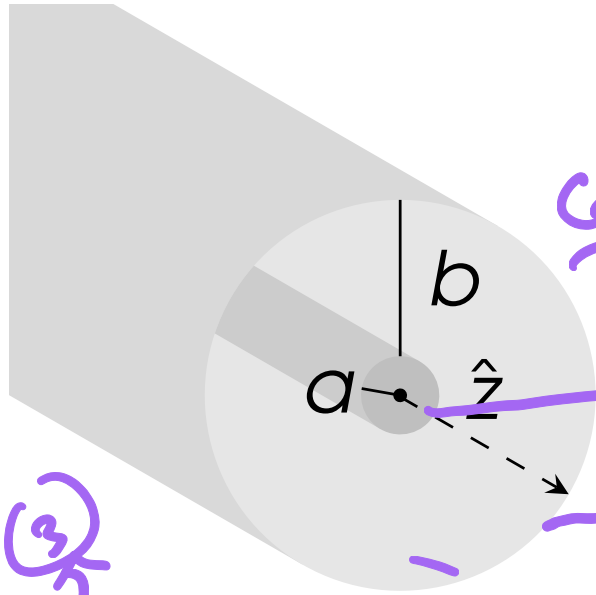
$Q_{encl.} = \int_0^L \int_0^{2\pi} \int_0^{\rho_0} \frac{Q_{TOT}}{\pi a^2 L} \rho \, d\rho \, d\phi \, dz$

$\frac{Q_{TOT}}{\pi a^2 L} (L)(2\pi)\left(\frac{\rho_0^2}{2}\right)$

$Q_{encl.} = \frac{Q_{TOT} \rho_0^2}{a^2}$

Put it all together: $2\pi\rho L E_\rho = \frac{Q_{TOT} \rho_0^2}{a^2} \frac{1}{\epsilon}$

Gauss' Law: Coaxial Cable (cont'd)



$$E_\rho = \frac{Q_{TOT}\rho_0^2}{2\pi a^2 \epsilon L \rho} \qquad \vec{E} = \hat{\rho} E_\rho$$

Cases:

① $0 < \rho < a \Rightarrow \rho_0 = \rho$

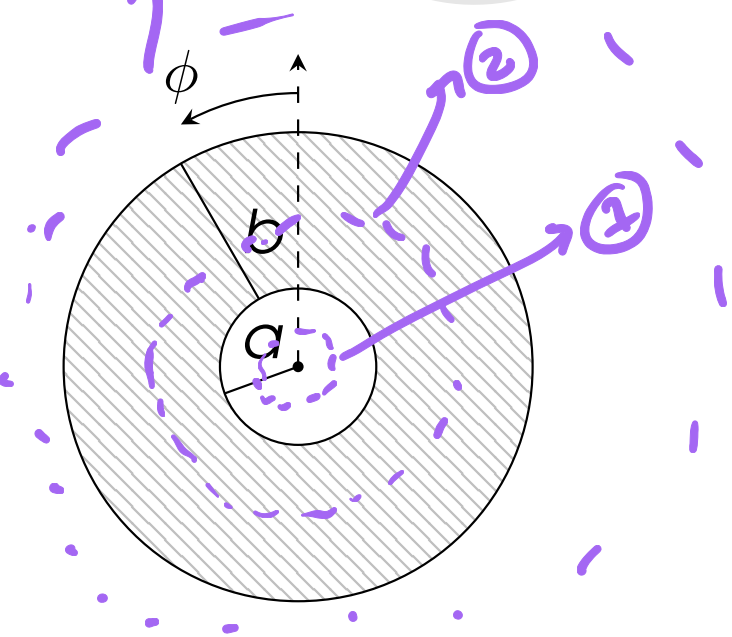
$$E_\rho = \frac{Q_{TOT}\rho^2}{2\pi a^2 \epsilon L \rho} = \frac{Q_{TOT}}{2\pi a^2 \epsilon L}\rho$$

② $a < \rho < b \qquad \rho_0 = a$

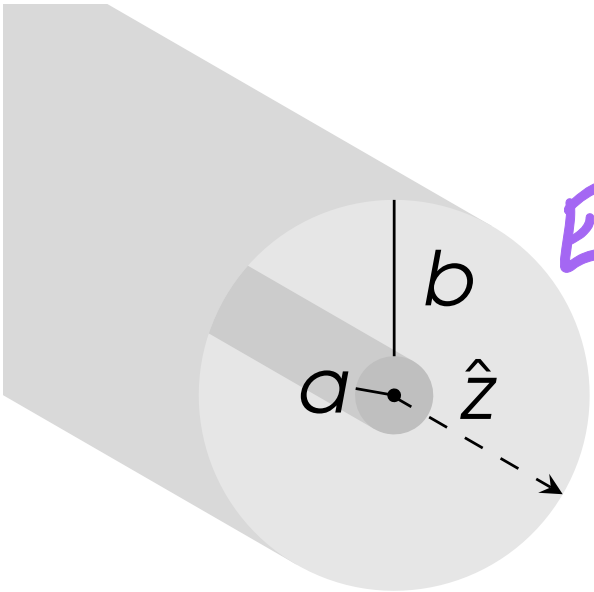
$$E_\rho = \frac{Q_{TOT}a^2}{2\pi\rho\epsilon L \rho} = \frac{Q_{TOT}}{2\pi\epsilon L}\left(\frac{1}{\rho}\right)$$

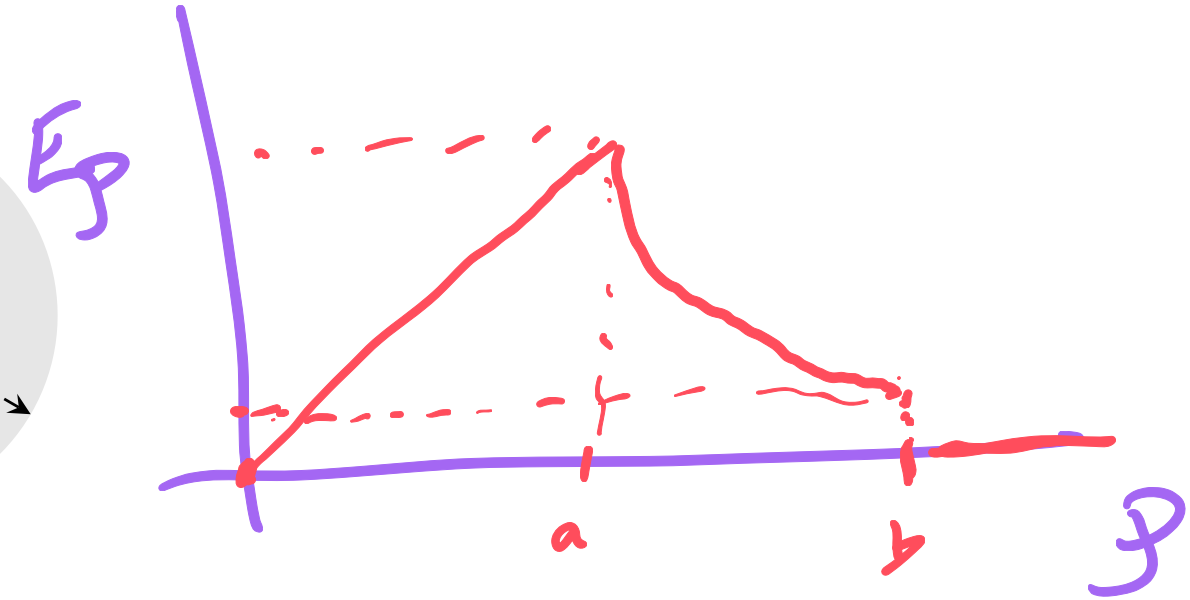
③ $\rho > b \qquad \rho_0 = a$

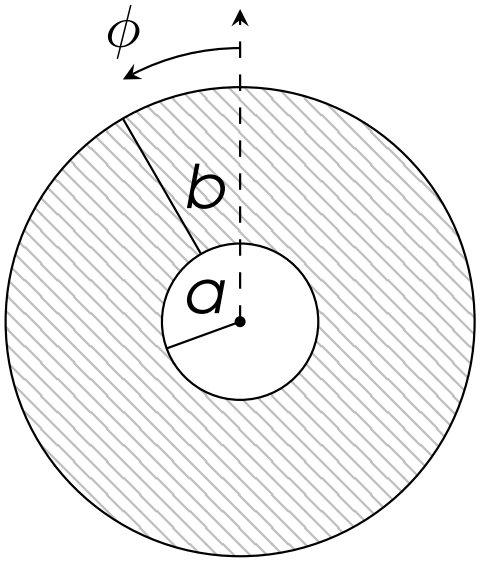
$Q_{TOT} = q^+ - q^- = 0 \Rightarrow E_\rho = 0$



Gauss' Law: Coaxial Cable (cont'd)







Gauss' Law: Point (differential) Form

Derivation

$$\oiint \vec{D} \cdot d\vec{S} = Q_{\text{encl}}$$

$$\iiint_V \nabla \cdot \vec{D} dV = \iiint_V \rho_V dV$$

$$\nabla \cdot \vec{D} = \rho_V$$

$$\nabla \cdot \vec{E} = \frac{\rho_V}{\varepsilon}$$

EXAMPLE

Given the following static electric field in free space, find the charge density (ρ) at the point $(2, -2, 0)$

$$\vec{E}(\vec{r}) = \hat{\mathbf{x}}Ax^2 + \hat{\mathbf{y}}Bz + \hat{\mathbf{z}}Cx^2z \longrightarrow A = 3\text{V}/\text{m}^3 \quad B = 2\text{V}/\text{m}^3 \quad C = 1\text{V}/\text{m}^4$$

Solution

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\varepsilon_0 \left[\frac{\partial}{\partial x}\left(3x^2\right) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}\left(x^2z\right) \right] = \rho$$

$$\varepsilon_0 \left(6x + 0 + x^2\right) = \rho$$

$$8.854 \times 10^{-12}\text{F/m}\left[(6)(2) + 2^2\right] = \rho$$

$$141.7\text{pC/m}^3 = \rho$$